Year 12 Maths - Coordinate geometry

Similar questions - Equation of a circle

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The equation $x^2 + y^2 - 4kx + 6y + 5k^2 = 0$, where k is a constant, can represent a circle.

(a) Show that the centre of the circle lies on the line $y = -3$ for every value of k, and find the radius in terms of k. **[4]**

(b) Find the range of values of k for which the equation represents a circle. **[2]**

(c) The circle touches the y-axis. Find the possible values of k, giving your answers in exact form. **[4]**

2 $A(-1,2)$ and $B(7,-4)$ are the endpoints of a diameter of a circle C.

(a) Find the coordinates of the centre of C and the exact radius of C. **[3]**

(b) Find the equation of C in the form $x^2 + y^2 + ax + by + c = 0$. **[2]**

(c) Show that the point $(0,3)$ lies on C. **[1]**

(d) Find the exact distance between the two points where C meets the x-axis. **[3]**

Question 1

(a) $(x-2k)^2+(y+3)^2 = 4k^2+9-5k^2 = 9-k^2$. Centre $(2k,-3)$, which always has $y=-3$. Radius $\sqrt{9-k^2}$

(b) Need $9-k^2>0$, so $-3<k<3$

(c) Touching the y-axis means radius $=|x$-coordinate of centre$|$: $9-k^2=4k^2$, so $k^2=\frac{9}{5}$ and $k=\pm\dfrac{3\sqrt{5}}{5}$ (accept $\pm\dfrac{3}{\sqrt{5}}$)

Question 2

(a) Centre is the midpoint: $(3,-1)$. Diameter $=\sqrt{8^2+6^2}=10$, so radius 5

(b) $(x-3)^2+(y+1)^2=25$, giving $x^2+y^2-6x+2y-15=0$

(c) $(0-3)^2+(3+1)^2=9+16=25$, so $(0,3)$ is on C

(d) Put $y=0$: $x^2-6x-15=0$, so $x=3\pm\sqrt{24}$. Distance $=2\sqrt{24}=4\sqrt{6}$